

1. Out of 275 students, 120 students study economics, 170 students study accounting, and 50 students study neither of the subjects. How many students do not study both subjects?

$$n(S) = 275, \quad n(E) = 120, \quad n(A) = 170$$

$$n(E' \cap A') = 50 = n(E \cup A)'$$

$$n(E \cup A) = n(S) - n(E \cup A)' = 275 - 50 = 225$$

$$n(E \cap A) = n(E) + n(A) - n(E \cup A) = 120 + 170 - 225 = 65$$

$$n(E \cap A)' = n(S) - n(E \cap A) = 275 - 65 = 210$$

2. If $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.3$, find $P(A \cup B)$, $P(A \cap B')$, and $P(A' \cup B')$.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.5 + 0.4 - 0.3 = 0.6$$

$$P(A \cap B') = P(A) - P(A \cap B) = 0.4 - 0.3 = 0.1$$

$$P(A' \cup B') = P(A \cap B)' = 1 - P(A \cap B) = 1 - 0.3 = 0.7$$

3. Consider a 4-sided dice and 2 coins are rolled at a time. Find the sample space of the outcomes. Also, find the probability of getting the same sides on the coins.

$$S = \left\{ \begin{array}{cccc} 1HH & 1HT & 1TH & 1TT \\ 2HH & 2HT & 2TH & 2TT \\ 3HH & 3HT & 3TH & 3TT \\ 4HH & 4HT & 4TH & 4TT \end{array} \right\} \rightarrow n(S) = 16$$

$$n(A) = n(\text{Getting the same sides on the coins}) = 8 \rightarrow P(A) = \frac{8}{16} = 0.5$$

4. Consider the trial on which the 6 is first observed in successive rolls of a six-sided die. Let A be the event that 6 is observed on the first trial. Let B be the event that at least two trials are required to observe the 6. Assuming sides are equal probable, find (a) $P(A)$, (b) $P(B)$, and (c) $P(A \cup B)$.

$$P(A) = \frac{1}{6}$$

$$P(B) = P(A') = 1 - P(A) = 1 - \frac{1}{6} = \frac{5}{6}$$

$$P(A \cap B) = 0$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{6} + \frac{5}{6} - 0 = 1$$

5. If A and B are mutually exclusive, find $P(A'|B)$ and $P(A' \cup B')$.

$$P(A'|B) = 1 - P(A|B) = 1 - 0 = 1$$

$$P(A' \cup B') = P(A \cap B)' = 1 - P(A \cap B) = 1 - 0 = 1$$

6. Given that $P(A \cup B) = 0.76$ and $P(A \cup B') = 0.87$, find $P(A)$.

$$P(A) = P(A \cup B') - P(A \cup B)' = P(A \cup B') - [1 - P(A \cup B)] = 0.87 - (1 - 0.76) = 0.63$$

7. An automobile company looks at its visitors and finds that (a) all purchase at least one car, (b) 80% purchase more than one car, (c) 30% purchase brand new cars, and (d) 15% purchase more than one brand new car. Find the probability that a visitor selected at random purchases exactly one car and it is not brand new.

$$P(\text{More than one car}) = P(A) = 0.8$$

$$P(\text{Brand new car}) = P(B) = 0.3$$

$$P(A \cap B) = 0.15$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.8 + 0.3 - 0.15 = 0.95$$

$$P(A' \cap B') = P(A \cup B)' = 1 - P(A \cup B) = 1 - 0.95 = 0.05$$

8. During a visit to a doctor's chamber, the probability of having neither a medical test nor a referral to a specialist is 15%. Of those coming to that chamber, the probability of having a medical test is 49%, and the probability of having a referral is 54%. What is the probability of having both a medical test and a referral?

$$P(\text{Having a medical test}) = P(A) = 0.49$$

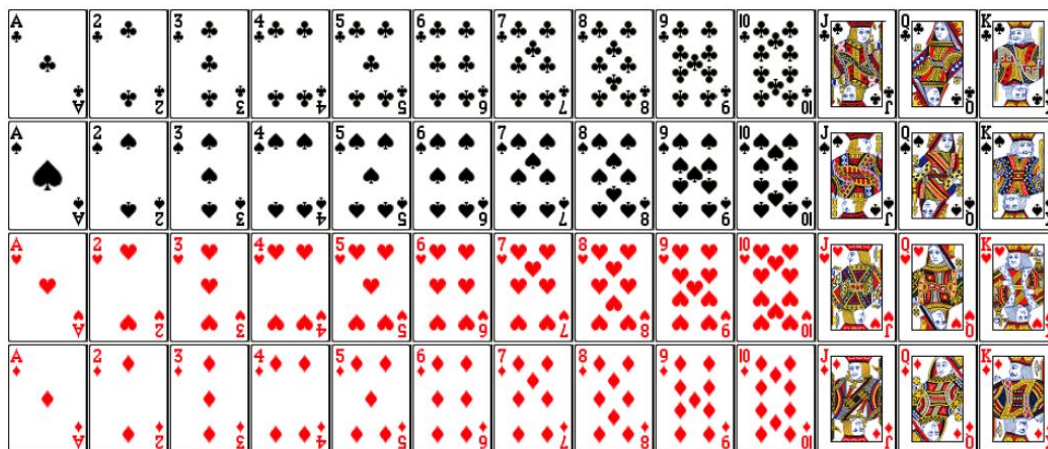
$$P(\text{Having a referral}) = P(B) = 0.54$$

$$P(A' \cap B') = P(A \cup B)' = 0.15$$

$$P(A \cup B) = 1 - P(A \cup B)' = 1 - 0.15 = 0.85$$

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.49 + 0.54 - 0.85 = 0.18$$

9. Draw one card at random from a standard deck of cards. The sample space S is the collection of the 52 equally probable cards. Assume that $A = \{x: x \text{ is a jack, queen, or king}\}$, $B = \{x: x \text{ is a 9, 10, or jack and } x \text{ is red}\}$, $C = \{x: x \text{ is a club}\}$, and $D = \{x: x \text{ is a diamond, a heart, or a spade}\}$. Find $P(A)$, $P(B)$, $P(A \cup B)$, $P(C)$, $P(D)$, and $P(C \cup D)$.



$$n(S) = 52, \quad n(A) = 12, \quad n(B) = 6, \quad n(C) = 13, \quad n(D) = 39$$

$$n(A \cap B) = 2 \text{ and } n(C \cap D) = 0$$

$$P(A) = \frac{12}{52} \text{ and } P(B) = \frac{6}{52}$$

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{2}{52}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{12}{52} + \frac{6}{52} - \frac{2}{52} = \frac{16}{52}$$

$$P(C) = \frac{13}{52} \text{ and } P(D) = \frac{39}{52}$$

$$P(C \cap D) = \frac{n(C \cap D)}{n(S)} = 0$$

$$P(C \cup D) = P(C) + P(D) - P(C \cap D) = \frac{13}{52} + \frac{39}{52} - 0 = 1$$

10. Two cards are drawn successively and without replacement from an ordinary deck of playing cards. Compute the probability of drawing a black ace on the first draw and a red card in the second draw.

$$\frac{2}{52} \times \frac{26}{51} = \frac{1}{51} = 0.0196$$

11. Paper is often tested for “burst strength” and “tear strength.” Say we classify these strengths as low, middle, and high. Then, after examining 100 pieces of paper, we find the following:

	A_1 (low)	A_2 (middle)	A_3 (high)	Total
B_1 (low)	7	11	13	31
B_2 (middle)	11	21	9	41
B_3 (high)	12	9	7	28
Total	30	41	29	100

$$P(A_1) = \frac{n(A_1)}{n(S)} = \frac{30}{100}$$

$$P(B'_2) = 1 - P(B_2) = 1 - \frac{n(B_2)}{n(S)} = 1 - \frac{41}{100} = \frac{59}{100}$$

$$P(A_3 \cap B_2) = \frac{n(A_3 \cap B_2)}{n(S)} = \frac{9}{100}$$

$$P(A_2 \cup B_3) = \frac{n(A_2 \cap B_3)}{n(S)} = \frac{9+12+7+21+11}{100} = \frac{60}{100}$$

$$P(A_1|B_2) = \frac{n(A_1 \cap B_2)}{n(B_2)} = \frac{11}{41}$$

$$P(B_1|A_3) = \frac{n(B_1 \cap A_3)}{n(A_3)} = \frac{13}{29}$$

$$P(B'_3|A_2) = 1 - P(B_3|A_2) = 1 - \frac{n(A_2 \cap B_3)}{n(A_2)} = 1 - \frac{9}{41} = \frac{32}{41}$$

12. A certain test for identifying cancer was tested among 1 lac people; we can expect results similar to those given in the following table, with some missing values. If one of 1 lac people is selected randomly, complete the table, and find the following probabilities: $P(A'_1|B_2)$, $P(B'_1|A_2)$, and $P(A_2 \cup B_2)$.

	A_1 (Carry cancer)	A_2 (Do not carry cancer)	Totals
B_1 (Test positive)	4500	9500	14000
B_2 (Test negative)	500	85500	86000
Totals	5000	95000	100000

$$P(A'_1|B_2) = 1 - P(A_1|B_2) = 1 - \frac{n(A_1 \cap B_2)}{n(B_2)} = 1 - \frac{500}{86000} = \frac{85500}{86000}$$

$$P(B'_1|A_2) = 1 - P(B_1|A_2) = 1 - \frac{n(A_2 \cap B_1)}{n(A_2)} = 1 - \frac{9500}{95000} = \frac{85500}{95000}$$

$$P(A_2 \cup B_2) = \frac{500+85500+9500}{100000} = \frac{95500}{100000}$$

13. At a sporting event, there are 30 non-replaceable balls on a table, of which 12 balls are yellow, 10 are red, and 8 are green. A player randomly hits the balls to win a prize. If the first ball hit is green, find the probability of (i) the next ball is green, (ii) the next ball is not green, and (iii) the next ball is a different color. Explain your findings.

$$P(G_1 \cap G_2) = P(G_1)P(G_2|G_1) = \frac{8}{30} \times \frac{7}{29} = 0.064$$

$$P(G_1 \cap G'_2) = P(G_1)P(G'_2|G_1) = \frac{8}{30} \times \frac{22}{29} = 0.202$$

$$P(G_1 \cap Y_2) + P(G_1 \cap R_2) = P(G_1)P(Y_2|G_1) + P(G_1)P(R_2|G_1) = \frac{8}{30} \times \frac{12}{29} + \frac{8}{30} \times \frac{10}{29} = 0.202$$

So, the next ball is not green, and the next ball is any different color, which means the same consequence.

14. A researcher finds that, of 982 men who died in 2002, 221 died from some heart disease. Also, of the 982 men, 334 had at least one parent who had some heart disease. Of the latter 334 men, 111 died from some heart disease. A man is selected from the group of 982. Given that neither of his parents had any heart disease, find the conditional probability that this man died of some heart disease.

	Heart disease	No heart disease	Total
Parental connection	111	223	334
No parental connection	110	538	648
Total	221	761	982

$$P(\text{Heart disease}|\text{No parental connection}) = \frac{n(\text{Heart disease} \cap \text{No parental connection})}{n(\text{No parental connection})} = \frac{110}{648}$$

15. In a bus out of 48 passengers, 34 are men, and in another bus out of 56 passengers, 24 are women. Say a passenger is transferred from the first bus to the second bus. For the capacity problem, one of the passengers of the second bus has to get off later. Find the probability that this passenger is a man.

$$n(\text{Total on the 1st bus}) = 48 \rightarrow n(M_1) = 34, n(W_1) = 14$$

$$n(\text{Total on the 2nd bus}) = 46 \rightarrow n(M_2) = 32, n(W_2) = 24$$

$$P(M_2|2\text{nd bus}) = P(M_1)P(M_2|M_1) + P(W_1)P(M_2|W_1) = \left(\frac{34}{48}\right)\left(\frac{32}{57}\right) + \left(\frac{14}{48}\right)\left(\frac{32}{57}\right) = \frac{1570}{2736} = 0.5738$$

16. A fruit shop contains 8 oranges and 7 apples. Another fruit shop contains an unknown number of oranges and 11 apples. A fruit is collected from each shop at random, and the probability of getting two fruits is the same is $\frac{149}{300}$. How many oranges were in the second shop?

$$P(O_1 O_2) + P(A_1 A_2) = \frac{149}{300} \rightarrow \frac{8}{15} \times \frac{x}{x+11} + \frac{7}{15} \times \frac{11}{x+11} = \frac{149}{300} \rightarrow \frac{8x+77}{15x+165} = \frac{149}{300} \rightarrow x = 11$$

17. Consider A and B be independent events with $P(A) = \frac{1}{4}$ and $P(B) = \frac{2}{3}$. Now, compute $P(A \cup B)$, $P(A \cap B')$, $P(A' \cap B)$, $P(A' \cap B')$, and $P(B|A)$.

$$P(A \cup B) = P(A) + P(B) - P(A) \times P(B) = \frac{1}{4} + \frac{2}{3} - \frac{1}{4} \times \frac{2}{3} = \frac{3}{4}$$

$$P(A)' = 1 - P(A) = 1 - \frac{1}{4} = \frac{3}{4}$$

$$P(B)' = 1 - P(B) = 1 - \frac{2}{3} = \frac{1}{3}$$

$$P(A \cap B') = P(A) \times P(B') = \frac{1}{4} \times \frac{1}{3} = \frac{1}{12}$$

$$P(A' \cap B) = P(A)' \times P(B) = \frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$$

$$P(A' \cap B') = P(A)' \times P(B') = \frac{3}{4} \times \frac{1}{3} = \frac{1}{4}$$

$$P(B|A) = \frac{P(A) \times P(B)}{P(A)} = P(B) = \frac{2}{3}$$

18. A survey was conducted among 200 university students to study the relationship between owning a laptop and subscribing to online learning platforms. The findings are summarized below:

Category	Own a laptop	Do not own a laptop	Total
Subscribe online	80	60	140
Do not subscribe online	20	40	60
Total	100	100	200

A is the event that a student owns a laptop; B is the event that a student subscribes to online learning. Then (i) find $P(A)$ and $P(B)$, (ii) test whether events A and B are independent.

$$P(A) = \frac{100}{200} = 0.5 \text{ and } P(B) = \frac{140}{200} = 0.7$$

$$P(A \cap B) = \frac{80}{200} = 0.4$$

$$P(A) \times P(B) = 0.5 \times 0.7 = 0.35 \neq 0.4 = P(A \cap B)$$

A and B are not independent.

19. Assume two 5-sided dice are rolled together and consider the events, $A = \{\text{odd sides on the first roll}\}$ and $B = \{\text{the sum of the dice is even}\}$. Determine whether the events are independent or not.

$$S = \left\{ \begin{matrix} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) \\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) \\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) \\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) \end{matrix} \right\} \rightarrow n(S) = 25$$

$$A = \{(1,1), (1,2), (1,3), (1,4), (1,5), (3,1), (3,2), (3,3), (3,4), (3,5), (5,1), (5,2), (5,3), (5,4), (5,5)\}$$

$$B = \{(1,1), (1,3), (1,5), (2,2), (2,4), (3,1), (3,3), (3,5), (4,2), (4,4), (5,1), (5,3), (5,5)\}$$

$$A \cap B = \{(1,1), (1,3), (1,5), (3,1), (3,3), (3,5), (5,1), (5,3), (5,5)\}$$

$$n(A) = 15 \rightarrow P(A) = \frac{15}{25} = 0.6, \quad n(B) = 13 \rightarrow P(B) = \frac{13}{25} = 0.52$$

$$n(A \cap B) = 9 \rightarrow P(A \cap B) = \frac{9}{25} = 0.36$$

No independent for $P(A) \times P(B) = 0.6 \times 0.52 = 0.31 \neq 0.36 = P(A \cap B)$

20. Two six-sided dice of red and white colors are rolled. The event A is such that the sum of the two outcomes is 7. The event B is such that the product of the two outcomes is 12. Find the probability of $A \cap B$ and $A \cup B$. Determine whether the events A and B are independent.

$$n(S) = 36, \quad n(A) = 6, \quad n(B) = 4, \quad n(A \cap B) = 2$$

$$n(A \cup B) = n(A) + n(B) - n(A \cap B) = 6 + 4 - 2 = 8$$

$$P(A) = \frac{6}{36}, \quad P(B) = \frac{4}{36}, \quad P(A \cap B) = \frac{2}{36}, \quad P(A \cup B) = \frac{8}{36}$$

$$P(A) \times P(B) = \frac{6}{36} \times \frac{4}{36} = \frac{2}{108} \neq P(A \cap B), \quad A \text{ and } B \text{ are not independent.}$$

21. Flip a fair coin twice and consider the events $A = \{\text{head in the first flip}\}$, $B = \{\text{head in the second flip}\}$, and $C = \{\text{the two flips yield different results}\}$. Determine whether the events A , B , and C are mutually independent or not.

$$S = \{HH, HT, TH, TT\} \rightarrow n(S) = 4$$

$$A = \{HH, HT\} \rightarrow n(A) = 2, \quad B = \{HH, TH\} \rightarrow n(B) = 2, \quad C = \{HT, TH\} \rightarrow n(C) = 2$$

$$P(A) = P(B) = P(C) = \frac{1}{2}$$

$$A \cap B = \{HH\} \rightarrow n(A \cap B) = 1, \quad B \cap C = \{TH\} \rightarrow n(B \cap C) = 1, \quad A \cap C = \{HT\} \rightarrow n(A \cap C) = 1$$

$$P(A \cap B) = \frac{1}{4} = \frac{1}{2} \times \frac{1}{2} = P(A)P(B)$$

$$P(B \cap C) = \frac{1}{4} = \frac{1}{2} \times \frac{1}{2} = P(B)P(C)$$

$$P(A \cap C) = \frac{1}{4} = \frac{1}{2} \times \frac{1}{2} = P(A)P(C)$$

$$A \cap B \cap C = \emptyset \rightarrow n(A \cap B \cap C) = 0$$

$$P(A \cap B \cap C) = 0 \neq \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = P(A)P(B)P(C)$$

A , B , and C are pairwise independent, but not mutually independent.

22. A red die and a white die are rolled together. Let the events $A = \{4 \text{ on the red die}\}$, $B = \{\text{sum of dice is odd}\}$, $C = \{5 \text{ on the red die}\}$, and $D = \{\text{sum of dice is 11}\}$. Show that A and B are independent, but C and D are not independent.

$$n(S) = 36, n(A) = 6, n(B) = 18, n(C) = 6, n(D) = 2$$

$$n(A \cap B) = 3, n(C \cap D) = 1$$

$$P(A) = \frac{6}{36} = \frac{1}{6}, P(B) = \frac{18}{36} = \frac{1}{2}, P(A \cap B) = \frac{3}{36} = \frac{1}{12}$$

$$P(A \cap B) = \frac{1}{12} = \frac{1}{6} \times \frac{1}{2} = P(A)P(B), A \text{ and } B \text{ are independent.}$$

$$P(C) = \frac{6}{36} = \frac{1}{6}, P(D) = \frac{2}{36} = \frac{1}{18}, P(C \cap D) = \frac{1}{36}$$

$$P(C \cap D) = \frac{1}{36} \neq \frac{1}{6} \times \frac{1}{18} = P(C)P(D), C \text{ and } D \text{ are not independent.}$$

23. If 3 men try to work randomly with the probabilities of success 0.5, 0.3, and 0.6, respectively. What is the probability that the work will be done? Also, find the probability that all of them work together.

$$P(\text{The work will be done}) = P(\text{At least one does the work})$$

$$= 1 - P(\text{No one does the work}) = 1 - (1 - 0.5)(1 - 0.3)(1 - 0.6) = 0.86$$

$$P(\text{All of them work together}) = (0.5)(0.3)(0.6) = 0.09$$

24. Suppose that A , B , and C are mutually independent events and that $P(A) = 0.5$, $P(B) = 0.8$, and $P(C) = 0.9$. Find the probabilities that (a) all three events occur, (b) exactly two of the three events occur, (c) exactly one of the three events occur, (d) none of the events occur, and (e) at least one of the events occurs.

$$P(A)' = 1 - P(A) = 1 - 0.5 = 0.5$$

$$P(B)' = 1 - P(B) = 1 - 0.8 = 0.2$$

$$P(C)' = 1 - P(C) = 1 - 0.9 = 0.1$$

$$P(\text{All three events occur}) = P(A)P(B)P(C) = 0.5 \times 0.8 \times 0.9 = 0.36$$

$$P(\text{Exactly two events occur}) = P(A)P(B)P(C)' + P(A)P(B)'P(C) + P(A)'P(B)P(C)$$

$$= 0.5 \times 0.8 \times 0.1 + 0.5 \times 0.2 \times 0.9 + 0.5 \times 0.8 \times 0.9 = 0.49$$

$$P(\text{Exactly one events occur}) = P(A)P(B)'P(C)' + P(A)'P(B)P(C)' + P(A)'P(B)'P(C)$$

$$= 0.5 \times 0.2 \times 0.1 + 0.5 \times 0.8 \times 0.1 + 0.5 \times 0.2 \times 0.9 = 0.14$$

$$P(\text{None of the events occur}) = P(A)'P(B)'P(C)' = 0.5 \times 0.2 \times 0.1 = 0.01$$

$$P(\text{At least one of the events occurs}) = 1 - P(\text{None of the events occur}) = 1 - 0.01 = 0.99$$

25. A factory has two machines producing the same metal bolt: Machine *A* produces 60% of the bolts with a defect rate 3%, whereas Machine *B* produces 40% of the bolts with a defect rate 5%. You pick one bolt at random from the daily production. What is the probability that the chosen bolt is defective? If the chosen bolt is found to be defective, what is the probability that it came from Machine *B*?

$$\begin{aligned}P(A) &= 0.6 \rightarrow P(D|A) = 0.03, & P(B) &= 0.4 \rightarrow P(D|B) = 0.05 \\P(D) &= P(A)P(D|A) + P(B)P(D|B) = 0.6 \times 0.03 + 0.4 \times 0.05 = 0.038 \\P(B|D) &= \frac{P(B \cap D)}{P(D)} = \frac{0.4 \times 0.05}{0.038} = 0.526\end{aligned}$$

26. A hospital receives 40% of its flu vaccine from Company *A* and 60% from Company *B*. Each shipment contains a large number of vials of vaccine. From Company *A*, 3% of the vials are ineffective; from Company *B*, 2% are ineffective. A hospital randomly selects vials from one shipment and finds that they are ineffective. What is the conditional probability that this shipment does not come from Company *B*?

$$\begin{aligned}P(A) &= 0.4 \rightarrow P(I|A) = 0.03, & P(B) &= 0.6 \rightarrow P(I|B) = 0.02 \\P(B'|I) &= 1 - P(B|I) = P(A|I) = \frac{P(A)P(I|A)}{P(A)P(I|A) + P(B)P(I|B)} = \frac{0.4 \times 0.03}{0.4 \times 0.03 + 0.6 \times 0.02} = 0.5\end{aligned}$$

27. In a company, two managers are working with a workload ratio of 2: 3. The managers make some errors, with 7% and 5% of their work. An investigating team found an error in the yearly work summary of the company. Find which manager is who is responsible for this incident?

$$\begin{aligned}P(M_1) &= 2x \rightarrow P(E|M_1) = 0.07, & P(M_2) &= 3x \rightarrow P(E|M_2) = 0.05 \\P(M_1|E) &= \frac{(2x)(0.07)}{(2x)(0.07) + (3x)(0.05)} = \frac{0.14}{0.29} = 0.48 \\P(M_2|E) &= \frac{(3x)(0.05)}{(2x)(0.07) + (3x)(0.05)} = \frac{0.15}{0.29} = 0.52\end{aligned}$$

So, 2nd manager will be responsible for the error.

28. A test indicates the presence of a particular disease 92% of the time when the disease is present and the presence of the disease 7% of the time when the disease is not present. If 5% of the population has the disease, calculate the conditional probability that a person selected at random has the disease if the test indicates the presence of the disease. Also, calculate the conditional probability that a person selected at random does not have the disease.

$$\begin{aligned}P(D^+) &= 0.05, P(D^-) = 0.95 \\P(T^+|D^+) &= 0.92 \rightarrow P(T^-|D^+) = 0.08, & P(T^+|D^-) &= 0.07 \rightarrow P(T^-|D^-) = 0.93 \\P(D^+|T^+) &= \frac{P(D^+) P(T^+|D^+)}{P(D^+) P(T^+|D^+) + P(D^-) P(T^+|D^-)} = \frac{0.05 \times 0.92}{0.05 \times 0.92 + 0.95 \times 0.07} = 0.409 \\P(D^-|T^+) &= 1 - P(D^+|T^+) = 1 - 0.409 = 0.591\end{aligned}$$

29. In a shopping mall, 3 customer executives sold 30%, 45%, and 25% of the total selling product in a financial year. The rates of efficiency of these three executives are 98%, 99%, and 96%, respectively. If a fault is found, find the probability that it may occur by the second executive.

$$P(E_1) = 0.3 \rightarrow P(F|E_1) = 1 - 0.98 = 0.02$$

$$P(E_2) = 0.45 \rightarrow P(F|E_2) = 1 - 0.99 = 0.01$$

$$P(E_3) = 0.25 \rightarrow P(F|E_3) = 1 - 0.96 = 0.04$$

$$P(E_2|F) = \frac{P(E_2) P(F|E_2)}{P(E_1) P(F|E_1) + P(E_2) P(F|E_2) + P(E_3) P(F|E_3)} = \frac{0.45 \times 0.01}{0.3 \times 0.02 + 0.45 \times 0.01 + 0.25 \times 0.04} = 0.22$$

30. At an office, officials are classified, and 30% of them are efficient, 50% are moderate, and 20% are unfit workers. Of the efficient ones, 15% left the job; of the moderate workers, 20% left the job, and of the unfit workers, 5% left the job. Given that an employee left the job. What is the probability that the employee is efficient or unfit? Also, find the probability that the employee is moderate.

$$P(E) = 0.3, \quad P(M) = 0.5, \quad P(U) = 0.2$$

$$P(L|E) = 0.15, \quad P(L|M) = 0.2, \quad P(L|U) = 0.05$$

$$P(E \cup U|L) = \frac{P(E) P(L|E) + P(U) P(L|U)}{P(E) P(L|E) + P(M) P(L|M) + P(U) P(L|U)} = \frac{0.30 \times 0.15 + 0.20 \times 0.05}{0.30 \times 0.15 + 0.50 \times 0.20 + 0.20 \times 0.05} = 0.36$$

$$P(M|L) = 1 - P(E \cup U|L) = 1 - 0.36 = 0.64$$